

EEG-based epileptic seizure detection using deep learning techniques: A survey

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ABSTRACT

Epilepsy is a complex neurological disorder marked by recurrent seizures, often stemming from abnormal discharge of the brain. Electroencephalogram (EEG) captures temporal and spatial shifts in cerebral electrical activity, holding pivotal diagnostic and therapeutic value for epilepsy. Deep learning techniques have made remarkable progress in EEG-based seizure detection over recent years. This review is dedicated to exploring seizure detection approaches based on deep learning, focusing on three distinct avenues. Primarily, we delve into the application of canonical deep learning methods in epilepsy detection. Subsequently, a more in-depth study was conducted on the hybrid models of deep learning. Next, the third is the integration of deep learning and traditional machine learning strategies. Finally, the challenges and future prospects related to this topic are put forward. The uniqueness of this review lies in its novel and comprehensive perspective on the latest research on deep learning-based epilepsy detection by systematically classifying methods, visualizing research progress, and addressing challenges and gaps in current research. It can provide valuable guidance for researchers who want to delve into the field of epileptic seizure detection based on EEG signals.

1. Introduction

Epilepsy is a disorder arising from cerebral functional impairments in affected individuals [1]. According to the latest research from the World Health Organization, approximately 65 million people worldwide are afflicted by epilepsy. Epileptic seizures can manifest across all age groups, with a higher incidence observed in infants and the elderly [2]. Epileptic seizures typically manifest with involuntary aberrant movements and instantaneous loss of consciousness, occasionally leading to fatality, thereby posing significant threats to an individual's well-being and quality of life [3,4]. The timely identification of epileptic episodes holds undeniable significance in alleviating potential adverse consequences.

A wealth of evidence indicates that the leveraging of electroencephalograms (EEGs) for diagnosing and detecting epileptic seizures has garnered significant favour among medical professionals [5–7]. EEG, as a means of measuring non-stationary signals of brain electrical activity,

offers a depiction of voltage fluctuations generated by neuronal ion currents in the brain. It's a time-consuming process that demands substantial effort from medical practitioners by observing EEG data. This complexity arises from the need to analyse intricate EEG data and identify subtle patterns indicative of epileptic activity. The manual scrutiny required to distinguish relevant signal changes from background noise, along with the necessity for expert interpretation, makes this diagnostic work meticulous and time-consuming. Hence, the adoption of computer-assisted diagnostics holds paramount significance in streamlining this process.

It is noteworthy to explore the evolution of automated seizure detection approaches over time. Fig. 1 presents a timeline highlighting pivotal milestones in the development of automated seizure detection methods, shedding light on the transformative journey from early initiatives to the integration of state-of-the-art deep learning techniques. Directing focus towards the historical trajectory of automated seizure detection methods, we can trace the origins to the pioneering

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endeavours of Gotman, who first proposed algorithms for the automatic detection of epileptic episodes [8]. Subsequent advancements led to Ottman and Qu refining the approach by integrating extended temporal contexts and harnessing the similarities between brainwave patterns [9]. In 1998, S. Park et al. designed an algorithm for long-term EEG seizure detection in epilepsy using an Artificial Neural Network (ANN) with wavelet transform preprocessing [10]. Subasi and Ercelebi employed neural networks and logistic regression to classify EEG signals [11], and M. Saab presented a system for detecting the onset of epilepsy seizure patterns in scalp EEG in 2005 [12]. In 2008, R. Meier et al. introduced a novel procedure for real-time automatic detection of various ictal patterns in human long-term EEG recordings, showcasing its accuracy and potential for early seizure detection [13].

While these early initiatives marked important milestones, the field continued to expand, witnessing the emergence of a plethora of signal processing techniques tailored for EEG analysis. In short, a comprehensive process for automated epileptic seizure detection primarily involves data collection and preparation, EEG preprocessing, feature extraction, and the classification of EEG signals (See Fig. 2). Among these stages, feature extraction and classification are pivotal steps, at times regarded as a unified process. Over time, an increasing number of machine learning-based approaches emerged, leveraging various methodologies to enhance the accuracy and reliability of seizure detection.

Efforts have been made to extract meaningful features from EEG signals using methods combining time-domain, frequency-domain, and time-frequency-domain approaches [14–18]. However, many of these methods necessitate meticulous manual parameter adjustments, leading

to a time-consuming process heavily reliant on empirical expertise. Challenges encompass decisions such as channel selection, appropriate decomposition levels in techniques like discrete wavelet transform (DWT) [19,20] or tunable-Q factor wavelet transform (TQWT) [21–23], and determining the intrinsic mode numbers in empirical mode decomposition (EMD) or variational mode decomposition (VMD) etc. Additionally, the selection of traditional machine learning (ML) techniques, including decision trees [24], random forests [25], support vector machines (SVM) [26–28], ensemble learning [29], bayesian linear discriminant analysis (BLDA) [30–32], and others, entails a trial-and-error approach. In general, the process of feature and classifier selection in traditional machine learning approaches involves trial and error method to some extent [33].

Despite traditional machine learning methods offer certain advantages, it's crucial to recognize their inherent limitations. These methodologies frequently necessitate domain expertise and manual intervention for parameter optimization, making them labor-intensive and susceptible to biases. The necessity of human involvement in tasks like feature selection and hyperparameter tuning drives us to explore alternative pathways for automated seizure detection, aiming to diminish reliance on manual input and enhance detection precision. In light of these considerations, the emergence of deep learning (DL) techniques has provided a promising pathway to address the challenges posed by traditional methods.

Numerous researches have been undertaken thus far to employ deep learning networks for the detection of epileptic seizures. As early as 2014, Turner et al. employed deep belief networks to realize seizure detection using multi-channel EEG data [34]. In 2016, Cilasun M.

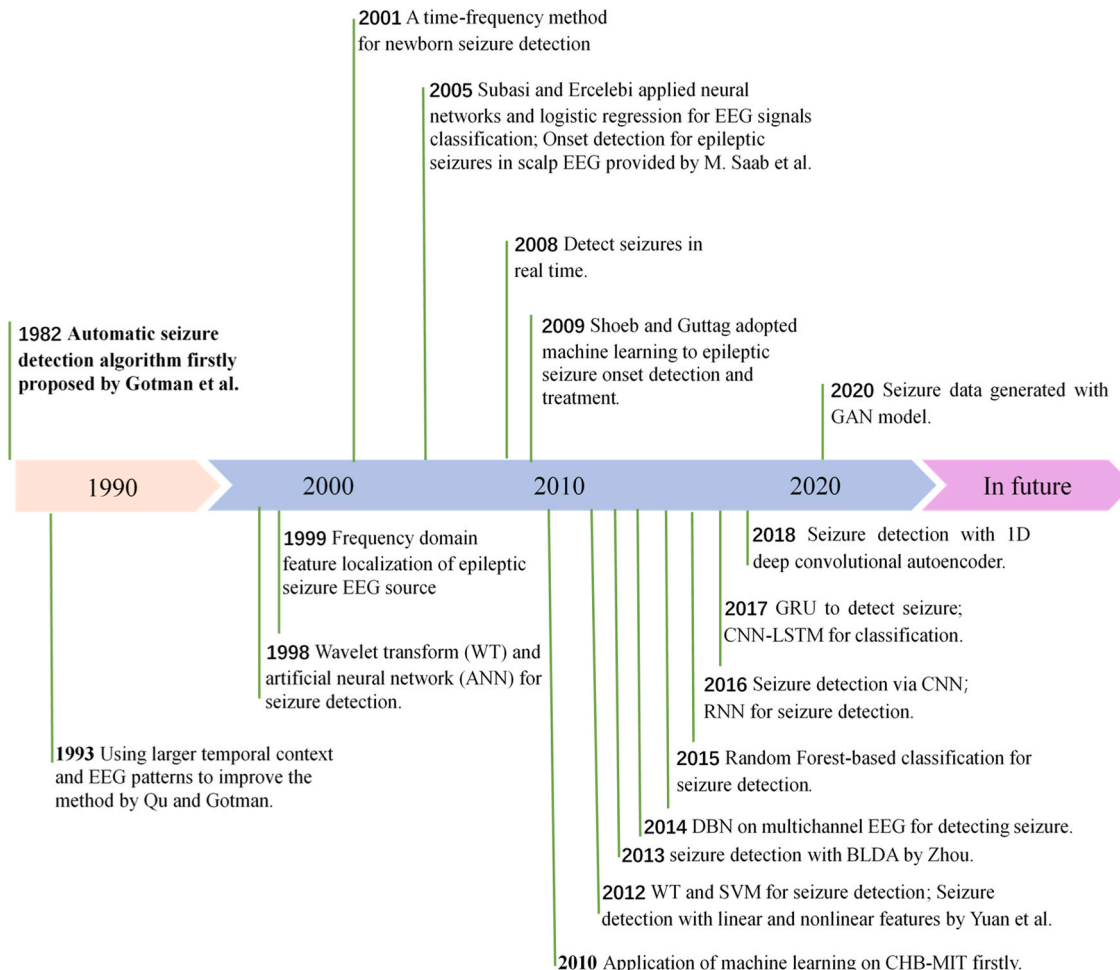


Fig. 1. A timeline for the development of measures for the automatic seizure detection by EEG.

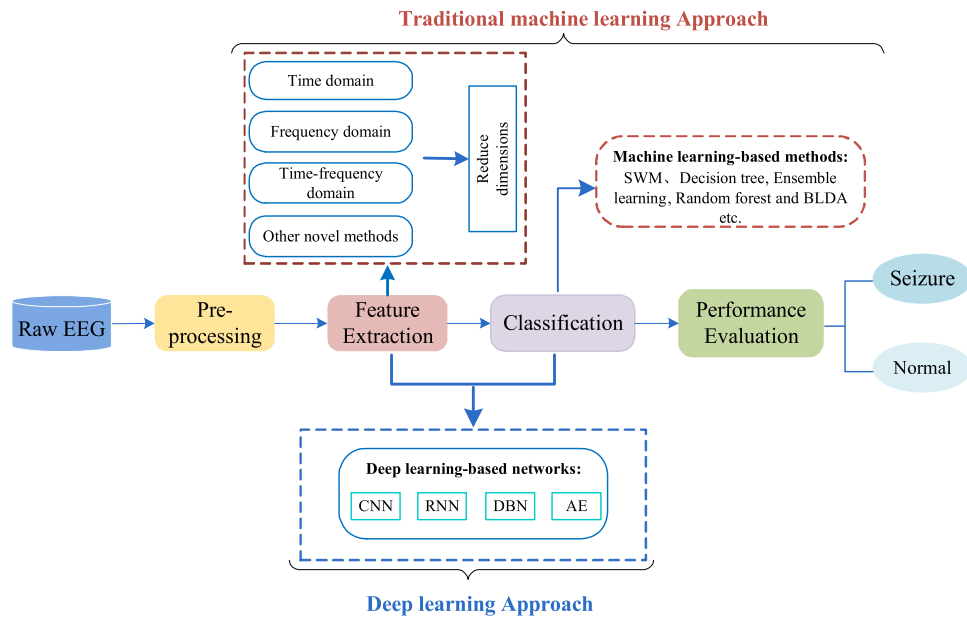


Fig. 2. The flowchart of a common seizure detection method. The top module is based on traditional machine learning methods, and the bottom module presents an end-to-end approach to classification based on deep learning networks.

focused on investigating the application of deep convolutional networks for seizure and non-seizure classification, achieving promising outcomes through experimentation on the Bonn dataset [35]. In the same year, Vidyaratne et al. provided a novel deep bidirectional recurrent neural network for EEG signal processing to enable seizure classification. Their study, conducted on 5 subjects from the CHB-MIT database, successfully detected 100 % of total seizure events [36]. Subsequently, in 2017, Talathi utilized a variant of recurrent neural networks, known as Gated Recurrent Unit (GRU), for seizure detection. Taking the Bonn dataset for evaluation, they achieved an overall accuracy approaching 100 % [37]. Golmohammadi et al. introduced a unique approach, presenting a hybrid architecture that integrates Convolutional Neural Network (CNN) and Long Short-Term Memory (LSTM) network for automated EEG classification [38]. The following year, Abdelhameed created a deep convolutional autoencoder-based automatic seizure detection framework [39]. Pascual et al. firstly applied GAN to synthesize the seizure data of epilepsy patients and train the detection algorithm, which effectively solved the problem of insufficient training data [40]. Throughout these studies, deep learning methodologies have been meticulously explored to advance the domain of automated seizure detection. Compared to traditional machine learning techniques, deep learning models exhibit enhanced performance in discriminating between epileptic seizure and non-seizure signals. This improvement can be attributed to the inherent capability of DL models to automatically learn and extract intricate features from raw data, thereby circumventing the need for manual feature engineering.

The subsequent sections of this review will delve into the applications of deep learning in automated seizure detection, encompassing classical deep learning methods, strategies for integrating deep learning with traditional machine learning techniques, and a focus on hybrid deep learning models. Differing from previously published review articles, which predominantly inclined to systematic overviews of the application of deep learning techniques in this context, our review takes a comprehensive approach by categorizing and encompassing the wide array of existing deep learning methods adopted. This all-encompassing approach aims to provide a nuanced understanding of the diverse landscape of deep learning methodologies for seizure detection. In this sense, the main contributions of this review are as follows:

- **Canonical DL Methods:** Discuss the application of foundational deep learning techniques in seizure detection, showcasing their potential in capturing intricate patterns and contributing to accurate seizure identification.
- **Hybrid DL Methods:** This section concentrates on investigating the fusion of different deep learning networks, such as common combinations like CNN-ANN and CNN-Autoencoder. It enhances the accuracy and robustness of epileptic seizure detection by harnessing the distinct advantages of each model.
- **Integrating DL with Traditional ML Methods:** This section underscores the exploration of synergy between deep learning and traditional machine learning approaches to further enhance the performance of epilepsy detection.
- **Challenges and Future Prospects:** Address the existing challenges and outline the potential future directions for further advancements in EEG-based seizure detection through deep learning methods.

The remaining sections of this review are presented as follows: [Section 2](#) offers an overview of commonly used EEG datasets for epileptic seizure detection. In [Section 3](#), various methods for signal preprocessing are introduced. The subsequent three sections delve into classical deep learning approaches, hybrid deep learning models, and the integration of DL with traditional ML methods, respectively. Discussions and challenges are addressed in [Section 7](#), while [Section 8](#) presents the summary and future prospects.

2. Seizure EEG dataset

At present, the EEG datasets used in research are collected intracranial or scalp EEG data. [Fig. 3](#) depicts various EEG datasets utilized in studies of epileptic seizure detection using deep learning techniques as referenced in this paper and their usage ratios. The pie chart proves that Bonn and CHB-MIT EEG datasets are used the most to validate deep learning models for automatic seizure detection. Moreover, nine prominent EEG datasets commonly adopted are shown in [Table 1](#) with comprehensive details.

3. EEG preprocessing

Due to the presence of artifacts and noise in EEG signals, the intrinsic

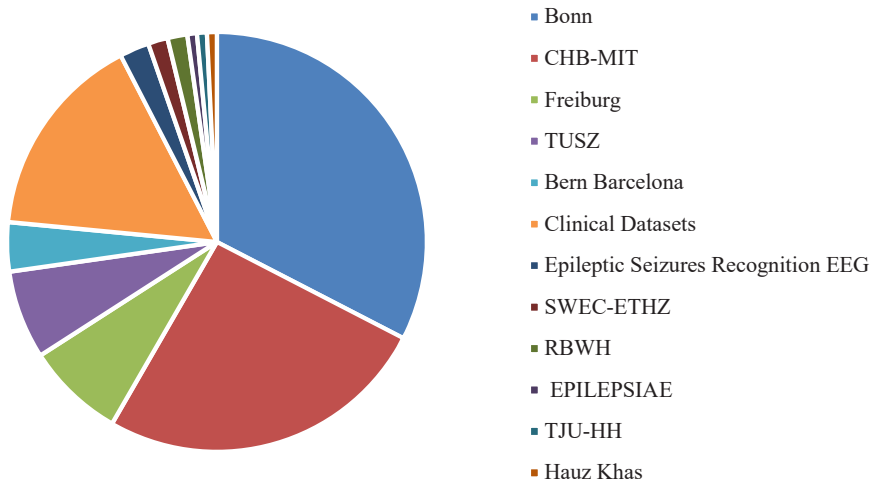


Fig. 3. The number of seizure detection models using DL techniques for each EEG dataset.

Table 1

Commonly used dataset for seizure detection.

Database	Number of patients	Sample frequency (Hz)	Times (hour)	Recording	Electrode type	URL
Bonn[41]	10	173.61	39 min	Surface and IEEG	Single-channel	https://www.ukbonn.de/epileptologie/?idcat=193!
Freiburg[42]	21	256	708	IEEG	Multi-channels	https://epilepsy.uni-freiburg.de/freiburg-seizure-prediction-project/eeg-database
CHB-MIT [43]	23	256	844	Scalp EEG	Multi-channels	https://physionet.org/content/chbmit/1.0.0/
Zenodo[44]	79	256	74 min	Scalp EEG	Multi-channels	https://zenodo.org/record/2547147
Bern Barcelona [45]	5	512/1024	83	IEEG	Multi-channels	https://www.upf.edu/web/ntsa/downloads/-/asset_publisher/xvT6E4pczrBw/content/2012-nonrandomness-nonlinear-dependence-and-nonstationarity-of-electroencephalographic-recordings-from-epilepsy-patients
Hauz Khas [46]	21	200	-	Scalp EEG	Multi-channels	https://www.researchgate.net/publication/308719109_EEG_Epilepsy_Datasets
Siena[47]	14	512	128	Scalp EEG	Multi-channels	https://physionet.org/content/siena-scalp-eeeg/1.0.0/
SWEC-ETHZ [48]	Long-term: 18 Short-term: 16	512/1024 512	2656 -	IEEG	Multi-channels Multi-channels	http://ieeg-swez.ethz.ch/
TUSZ[49]	307	250	-	Scalp EEG	Multi-channels	https://isip.piconepress.com/projects/tuh_eeg/html/downloads.shtml

information of the EEG signals may become compromised. Meanwhile, distinguishing epileptic seizure patterns from other EEG activities becomes challenging, thereby presenting a formidable challenge in the task of seizure detection. Consequently, prior to inputting the data into deep neural networks, preprocessing operations are imperative for EEG signals, which play a pivotal role in enhancing the quality of EEG data for subsequent analysis. For instance, Gaussian filtering is effective in removing high-frequency noise, while Principal Component Analysis (PCA) and Independent Component Analysis (ICA) are widely used to separate noise from the signal, such as ocular or muscular artifacts. Techniques like Empirical Mode Decomposition (EMD) and Variational Mode Decomposition (VMD) are beneficial in isolating intrinsic mode functions that capture relevant signal features for seizure detection.

In addition, a series of wavelet transform techniques have got broad application during the preprocessing stage of EEG signals, such as Wavelet Packet Decomposition (WPD), Discrete Wavelet Transform (DWT), Tunable Q-factor Wavelet Transform (TQWT), and so on [50]. The study by Liu proved that VMD offered notable superiority over DWT and EMD in distinguishing between focal and non-focal signals [51]. Alickovic et al. enhanced the representation of EEG signals by combining three different decomposition methods, i.e. EMD, DWT, and

WPD [52]. Furthermore, normalization techniques such as the Z-score method are employed to achieve data normalization in practice, ensuring that the input data conforms to a standard distribution, which is crucial for the stability and performance of deep learning models.

4. Canonical DL-based methods

This section explores canonical deep learning methodologies, encompassing powerful neural network architectures. Subsection 4.1 covers CNNs, renowned for hierarchical feature extraction and spatial translation invariance, ideal for image analysis. Subsection 4.2 delves into Recurrent Neural Networks (RNNs), including LSTM and GRU variants, which are suitable for capturing temporal dependencies in sequential data. Subsection 4.3 discusses Deep Belief Networks (DBNs) with stacked layers for feature learning. Subsection 4.4 presents Autoencoders, which is used for facilitating dimensionality reduction and latent space representation. Additionally, innovative applications of Graph Neural Networks (GNNs) and Generative Adversarial Networks (GANs) in seizure detection are explored. This section provides insights into these architectures' contributions to epileptic seizure detection. This comprehensive exploration of canonical DL-based methods

provides insights into their distinct architectures and contributions towards the field of epileptic seizure detection.

4.1. CNNs

As a category of feedforward neural networks, CNNs have exhibited remarkable proficiency in the domains of image analysis and feature extraction [53]. Initially recognized for their inherent capacity to capture intricate spatial hierarchies and identify complex patterns within image data, CNNs swiftly ascended to prominence as a prevailing architectural paradigm. So far, an increasing body of research has been devoted to their application in the realm of EEG, where their notable performance in distinguishing epileptic seizure signals from non-seizure signals has become particularly evident. Fig. 4 illustrates the typical workflow using 1D-CNNs.

In the study [54], authors employed a convolutional neural network and the Fast Fourier Transform (FFT) approach to detect epileptic seizures with raw EEG signals. The CNN outperforms manual feature extraction methods, achieving high accuracy in distinguishing seizure states across various experiments using both time and frequency domain signals. Frequency domain signals exhibit superior performance and potential for CNN applications compared to time domain signals. Acharya et al. successfully developed a 13-layer deep CNN structure for classifying normal, interictal, and ictal EEG segments. They evaluated the model on Bonn dataset, achieving an accuracy of 88.7 % [35]. Spectrograms describe the relationship between signal frequency and amplitude, commonly utilized for analyzing signal characteristics. Yan et al. constructed four CNN models based on VGG-net architecture to autonomously detect epileptic activities in EEG spectrogram analysis and evaluated their performance on CHB-MIT and NYP-WC databases, respectively [55]. Atal and Singh proposed using Greedy Particle Swarm Optimization (GPSO) to optimize extracted features in the time and frequency domains to select the most optimal features. A novel CNN network is then used to classify these optimized features. The distinctive point of this study lies in the introduction of a modified Blackman bandpass filter (MBBF) to discriminate signals passing through specific frequency ranges while attenuating those outside the designated range [56].

The aforementioned studies are all explorations based on 1D-CNNs, where EEG signals serve as inputs to convolutional networks in one-dimensional form. However, compared to 1D-CNNs, 2D-CNNs hold greater potential for EEG signal analysis. Despite the advantage of having fewer parameters, the classification performance of 1D-CNNs may not match that of 2D-CNNs in certain scenarios. There are studies have shown that 2D-CNNs could better capture spatial dependencies between different EEG electrodes, leading to improved accuracy in seizure detection tasks. Additionally, 2D-CNNs have been successfully applied to spectrograms and other 2D representations of EEG signals. In 2019, Türk and Özerdem used Continuous Wavelet Transform (CWT) to evaluate the frequency time-scale map of the original EEG data directly in the 2D-CNN architecture and obtained effective results on the Bonn dataset [57]. Followed by next year, Jing et al. made a major breakthrough with the ability of 2D-CNN to automatically detect interictal

epileptiform discharges (IEDs) in EEG recordings [58]. Prasanth attaches the filtered signal vertically to form two-dimensional matrix, which was placed into the designed 2D-CNN model to realize the effective detection of epilepsy, which confirmed that 2D-CNN is more effective than 1D-CNN [59]. In [60], the authors presented a 2D-CNN framework, where the input data has one dimension representing time steps, and the other dimension represents EEG electrodes. This model extracts spectral and spatial features from EEG signal recordings to identify epileptic seizures. The article [61] presents a novel approach involving empirical mode decomposition for generating initial EEG features, Weighted overlook Graphs (WOG) representation of IMFs after EMD decomposition, followed by classification using a 2D-CNN network. Table 2 lists nearly five years of some researches have used the CNNs and Fig. 5 shows a comparison of their accuracy rates. Indeed, there are studies applying 3D-CNNs in the field of epilepsy detection, exemplified by [62] and [63], albeit infrequently.

4.2. RNNs

RNNs, a fundamental class of neural architectures designed to excel in tasks involving sequential and temporal data analysis. RNNs have demonstrated their proficiency in capturing dependencies across sequences, making them particularly apt for time-series data, including EEG signals in epileptic seizure detection [36]. RNN has two variants LSTM and GRU which are helped is often applied in epilepsy detection field, so they are discussed respectively. A typical flow using RNNs is shown in Fig. 6.

4.2.1. LSTM

A specialized variant of RNNs, Long Short-Term Memory (LSTM) designed to mitigate the vanishing gradient problem and facilitate the capture of long-range dependencies by creating a gated mechanism. LSTMs have garnered significant attention due to their capacity to effectively model sequential patterns while retaining memory of information over extended sequences. Some investigations have shown that LSTM achieves positive results in both binary-classification (inter-ictal or ictal) and three-class classification (pre-ictal, inter-ictal and ictal) problems [69]. In the study of [70], an automatic seizure detection system based on LSTM was presented and achieved an average accuracy of 95.54 %. Aliyu and Lim proposed an LSTM classification model and employed a feature selection algorithm to choose the optimal wavelet features of EEG [71]. Bidirectional LSTM (BiLSTM) is also frequently employed. In [72], the pre-processed data obtained by z-score normalization and Savitzky-Golay filtering were used, and then sent into the pre-defined BiLSTM deep network to automatically identify seizure signals. The evaluation on the Bernard database yielded an accuracy of 99.6 %. Tuncer and Bolat initially extracted instantaneous frequency and spectral entropy as features, and directly feed them into BiLSTM classifier, emphasizing the significance of BiLSTM model hyper-parameters [73].

4.2.2. GRU

Another prominent variant of RNNs, the Gated Recurrent Unit (GRU)

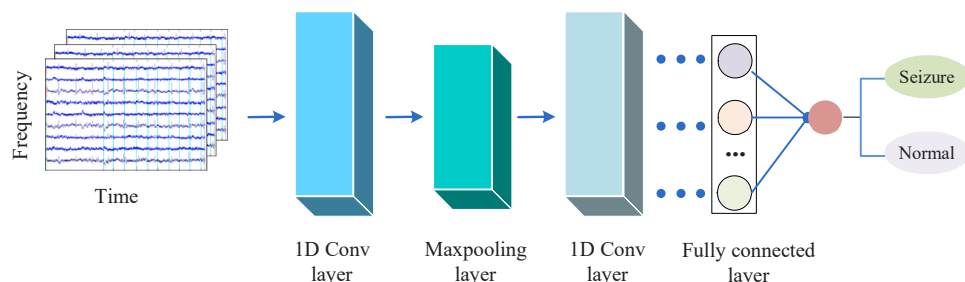
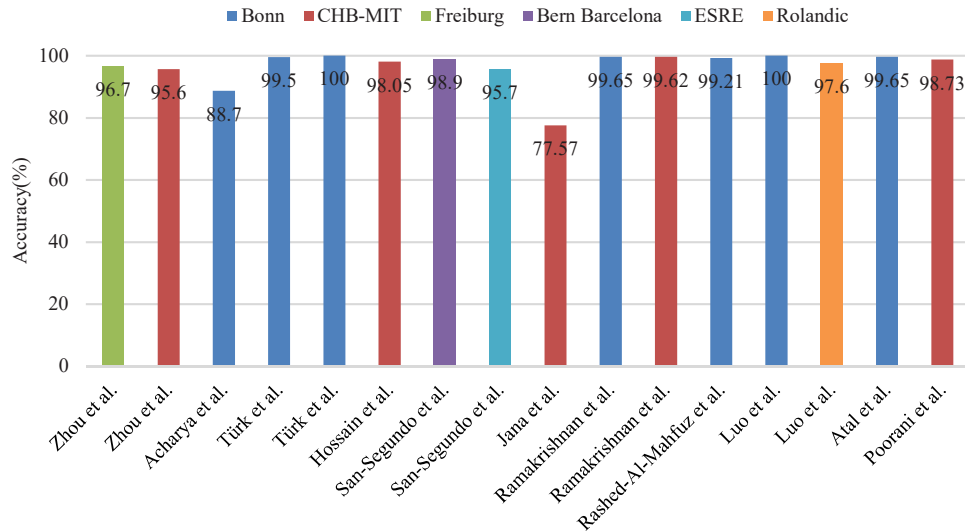
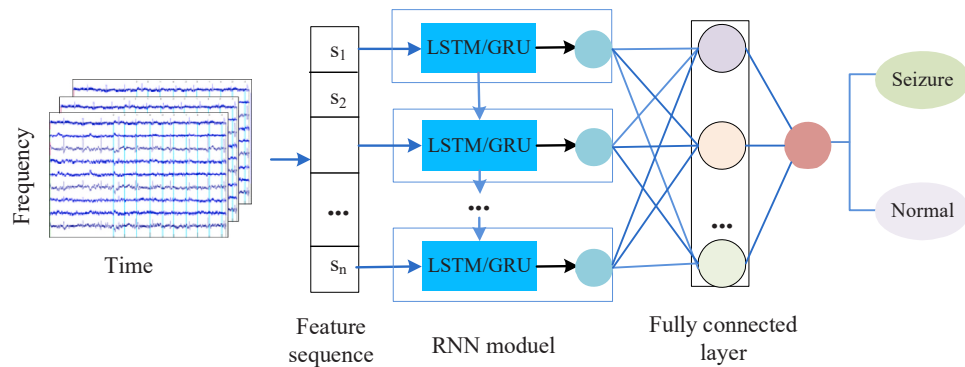


Fig. 4. The typical workflow using 1D CNNs for seizure detection.

Table 2

Existing literature on using CNNs (1D-CNN or 2D-CNN) in recent 5 years.

Work	Year	Preprocessing technique	Method	Database	Accuracy (%)
Zhou et al. [54]	2018	FFT	1D-CNN	Freiburg	96.7
Acharya et al. [35]	2018	Z-score normalization	13-layer deep 1D-CNN structure	CHB-MIT	95.6
Yan et al. [55]	2019	-	Spectrogram sampling + VGG-net-based CNN models	Bonn	88.7
Türk et al. [57]	2019	CWT	2D-CNN	CHB-MIT	-
Hossain et al. [60]	2019	-	Spectral, spatial, temporal features +2D CNN	NYP-WC	AE-BE:99.5
San-Segundo et al. [64]	2019	Fourier, wavelet, EMD transforms	2D-CNN	Bonn	AD-BD:100
Jana et al. [65]	2020	Butterworth filter	Spectrogram+1D-CNN	CHB-MIT	98.05
Ramakrishnan et al. [66]	2020	-	1D-AaLBP & 1D-AdLBP+ 1D-CNN	Bonn	98.9
Rashed-Al-Mahfuz et al. [67]	2021	STFT and CWT	2D-CNN	CHB-MIT	95.7
Luo et al. [61]	2022	EMD-WOG	2D-CNN	Bonn	77.57
Atal et al. [56]	2023	MBBF	10 time domain, 5 frequency domain feature with GPSO optimized + 1D-CNN	Bonn	99.65
Poorani et al. [68]	2023	Normalization	1D-CNN	Rolandic	99.62
				Bonn	99.21
				Bonn	100
				Bonn	97.6
				Bonn	99.65
				Bonn	98.73

**Fig. 5.** The comparison of accuracy attained by different authors using CNNs models for identifying epilepsy.**Fig. 6.** The typical workflow using RNNs for epileptic seizure detection.

designed to address the vanishing gradient issue and enhance the modelling of sequential dependencies. GRUs offer a streamlined architecture with simplified gating mechanisms, which balances the trade-off

between modelling capacity and computational complexity. The method was presented in [37], where the application of GRU in seizure detection is analysed, and the overall accuracy of the Bonn dataset is close to

100 %. The authors of [74] created a 5-layer gated recurrent unit neural network that was applied to EEG data after DWT decomposition with an accuracy of 98.5 %. Similarly, the research [75] proposed a 4-layer GRU network for automatic seizure detection, avoiding the extraction of any hand-made features. GRU and BiGRU networks were compared in [76] yielding accuracy rates of 98.6 % and 99.3 %, respectively, for binary classification on the Bonn dataset. Hence, it can be concluded that BiGRU exhibits slightly superior performance in recognition tasks compared to GRU networks. Bidirectional BiGRU networks were designed in the work of Zhang et al. The energy output after DWT is provided to BiGRU to distinguish epileptic and non-epileptic states [77]. Motivated from these recent accomplishments, it is evident that GRUs have made significant contributions in the field of epileptic signals detection. Table 3 covers nearly five years of studies about the RNNs-based architecture, which includes LSTMs or GRUs. The comparison plot of the accuracy (%) gained by different authors using RNNs networks for recognizing seizure activity is shown in Fig. 7.

4.3. DBNs

DBNs put forward by research [84], which are a distinctive category of neural networks renowned for their hierarchical and generative characteristics. DBNs have garnered considerable attention due to their capacity to harness the benefits of both unsupervised and supervised learning for feature extraction and representation learning [85]. Several studies have investigated the application of Deep Belief Networks in the identification of epileptic EEG signals. Le et al. extracted the wavelet features of EEG peaks in 2018 and then fed them into the DBNs network to perform binary classification [86]. A novel feature selection technique based on Oppositional Aquila Optimizer (OAOFs) is proposed to deal with the pre-processed EEG signals by the authors of [87]. The selected best features were sent to the DBNs structure with the Artificial Ecosystem Optimizer (AEO) for classification, and the optimal accuracy result was obtained at 97.81 %.

4.4. Autoencoders

Delving into the autoencoders (AEs), as invented by Hinton [88], it becomes evident that they primarily consist of two components: the

encoder and decoder. Autoencoder excels in learning compressed representations of input data. denoising AE (DAE), stacked AE (SAE), and sparse AE (SpAE) are common variants of AE. It acquires initial layer weights through layer-wise pretraining, facilitating feature dimension reduction and efficient data encoding. Thus, research on AEs frequently emerges in the field of EEG signal processing [89–97].

Yu et al. combined unsupervised feature learning to build a robust automatic detection system for identifying epilepsy from EEG data based on stacked AE, called R-SAE, in which the maximum coreentropy criterion (MCC) was used to reduce noise/outliers in EEG signals [98]. Another novel automatic detection procedure based on the stacked sparse autoencoder with a softmax classifier was provided by Qin et al. in 2016, which achieved an overall average accuracy of 96 % on the Bonn dataset [99]. In [100], a semi-supervised stacked AE was presented for the detection of epileptic events. In [91], it extends their work from [101] and presents an end-to-end epileptic seizure detection framework known as Wave2Vec, which employs CWT to generate intrinsic and temporal features, followed by using sparse autoencoders (SpAE). Testing on the CHB-MIT EEG database yields an accuracy of 93.92 %. Compared to [101], this research demonstrates that softmax is more suitable for detection tasks. The literatures on AEs for seizure detection is presented in Table 4 and Fig. 8.

4.5. GNNs

GNN is a type of neural network architecture designed for processing and learning from graph-structured data. The core idea behind GNNs is to iteratively update node representations by aggregating information from neighboring nodes, enabling them to capture the structural properties and relationships within the graph [102]. Numerous GNN variants, including Graph Convolution Network (GCN) and Graph Attention Network (GAT), have been applied in the domain of epilepsy detection. These GNNs empower researchers to represent brain connectivity data as graphs, where nodes typically represent EEG electrodes and edges indicate the strength of the connections between them or the dependence in different brain regions. This graph-based representation captures the spatial relationships.

The construction of the adjacency matrix is a fundamental aspect of graph data representation, particularly vital for comprehending

Table 3
Existing literature on using RNNs in recent 5 years.

Work	Year	Preprocessing technique	Method	Database	Accuracy (%)
Ahmedt-Aristizabal et al. [70]	2018	-	LSTM	Bonn	95.54
Abbasi et al. [69]	2019	Discrete Cosine Transform	Hurst & ARMA features +LSTM	Bonn	99.17
Hussein et al. [78]	2019	Band-pass filter	LSTM	Bonn	100
Fraïwan et al. [72]	2020	Z-Score Normalization & Savitzky-Golay Filtering	BiLSTM	Bern-Barcelona	99.6
Hu et al. [79]	2020	Local Mean Decomposition	statistical features +BiLSTM	CHB-MIT	-
Aliyu et al. [71]	2021	DWT	20 eigenvalue features with optimal CCP-selected +LSTM	Bonn	98
Tuncer et al. [73]	2022	Min–Max Normalization	Instantaneous frequency & spectral entropy +BiLSTM	Bonn	99
Tuncer et al. [80]	2022	DWT	13 statistical features with CFS-selected +LSTM	TUSZ	98.08
Goel et al. [81]	2023	-	Entropy features +LSTM	Bern Barcelona	93.64
Fukumori et al. [82]	2019	Butterworth filter & DWT	GRU	Benign Epilepsy with Centro-Temporal Spikes	-
Verma et al. [74]	2021	DWT	5-layer GRU network	Bonn	98.5
Singh et al. [76]	2022	-	GRU	Bonn	98.6
			BiGRU		99.3
Zhang et al. [77]	2022	DWT	Energy value +BiGRU network	CHB-MIT	98.49
Ramwala et al. [75]	2023	-	4-layer GRU network	Bonn	97.5
Mekruksavanich et al. [83]	2023	Normalization	LSTM	Epileptic Seizures Recognition EEG	98.2
			BiLSTM		98.2
			GRU		98.83
			BiGRU		98.46

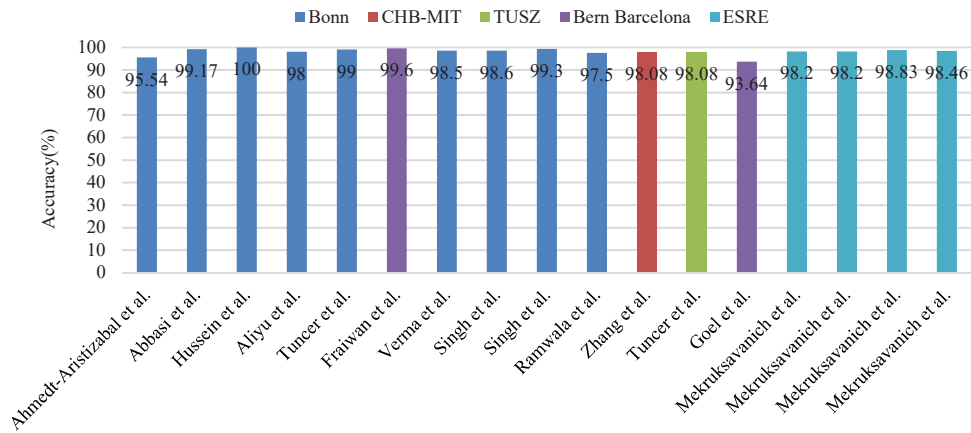


Fig. 7. Comparison of accuracy attained by different authors using RNNs models for identifying epilepsy.

Table 4

Existing literature using AEs in recent 5 years.

Work	Year	Preprocessing technique	Method	Database	Accuracy (%)
Gasparini et al. [92]	2018	CWT	Time-frequency map + SAE	Clinical Datasets	86.5
Karim et al. [93]	2018	DWT	SAE	Bonn	96
Sharathappriya et al. [94]	2018	Harmonic Wavelet Packet Transform	Katz-based Fractal dimension + AE	Bonn	98.67
Karim et al. [95]	2018	-	Taguchi Method + Stacked Sparse Autoencoder (SSAE)	Bonn	100
Yuan et al. [91]	2019	CWT	SpAE	CHB-MIT	93.92
Karim et al. [96]	2019	high-pass, low-pass and notch filtering	AE	Clinical Datasets	-
Emami et al. [97]	2019	Dimension reduction to one-dimensional data	Energy spectral density+ SAE	Clinical Datasets	100

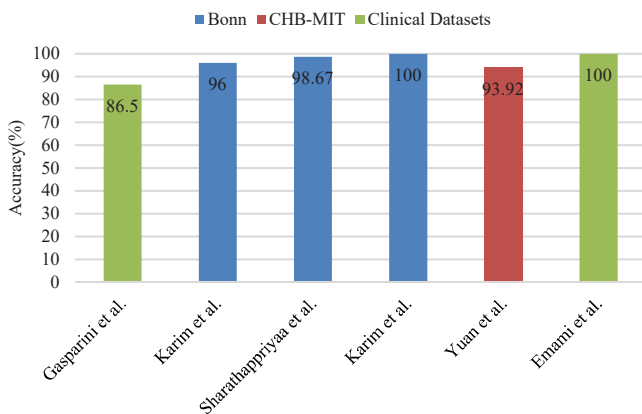


Fig. 8. Comparison of accuracy attained by different authors using AEs models for identifying epilepsy.

interconnections among nodes. The adjacency matrix can reflect coupling relationships among electroencephalogram electrodes. Researchers employ diverse methods to quantify electrode relationships for constructing the pivotal matrix. Here are some commonly used approaches: Pearson Correlation Coefficient (Calculating instantaneous linear interdependence between signals based on time-domain amplitude values, this method is widely applied); Phase Locking Value (PLV: measures phase synchronization degree between EEG electrode signals, focusing on phase coherence); Phase Lag Index (Like PLV, PLI assesses phase differences between signals, emphasizing phase distribution asymmetry); Wavelet Coherence (WC often viewed as a qualitative estimation, which characterizes dynamic signal relationships in the time-frequency domain). The selection of an appropriate adjacency

matrix construction method depends on epilepsy detection task specifics. Diverse ways offer varied relationship measurement granularity, aiding in uncovering electrode correlations and enhancing comprehension of brain activity patterns.

A temporal graph convolutional network (TGCN) model was designed in [103], where EEG signals were analysed by STFT preprocessing method. This model integrates the principles of GNN and time-series convolution models, effectively harnessing the temporal and spatial information from EEG data, tested on the data came from patients who received an epilepsy diagnosis at the Cleveland Clinic Foundation's Adult Epilepsy Monitoring Unit (EMU) between 2005 and 2017 within the brain, which is crucial for detecting abnormal patterns associated with seizures. In 2021, Zhao et al. introduced two end-to-end networks, Linear Graph Convolution Network (LGCN) [104] and Graph Attention Network [105], for the identification of epileptic activities in multi-channel EEG signals. In these work, focal loss is employed to address the issue of imbalanced seizure and non-seizure data. In Tang's work, a self-supervised Graph Neural Network is presented for the characterization and classification of EEG data in epilepsy patients [106]. Rahmani et al. provided a model based on GAT with meta-learning ideas for recognizing epilepsy [107]. Almost all recent studies involving GNNs-based epilepsy detection are presented in Table 5 and Fig. 9, respectively. It's worth noting that this table encompasses solely individual GNN or its modifications. Recently, some state-of-the-art hybrid model techniques based on Graph Neural Networks have emerged, and these methodologies will be extensively discussed in Section 5.

4.6. GANs

GANs, short for Generative Adversarial Networks, is a deep learning framework that typically serves unsupervised learning tasks and consists

Table 5

Existing literature on using GNNs (or its modifications) in recent 5 years.

Work	Year	Preprocessing technique	Method	Database	Accuracy (%)
Covert et al. [103]	2019	STFT	TGCN	EMU	-
Chen et al. [108]	2020	ICA	EGCN	Bonn CHB-MIT	99.8 98.35
Wang et al. [109]	2020	FFT	Frequency-domain complex network features + GCN	Bonn Private EEG (SSW)	100 89.3
Nhu et al. [110]	2021	Band-pass Filter	Different GCN architectures in IED detection	Clinical Datasets	-
Zhao et al. [104]	2021	Band-stop filter	LGCN +focal loss	CHB-MIT	99.3
Zhao et al. [105]	2021	Band-pass Filter & Z-score standardization	GAT +focal loss	CHB-MIT	98.89
Tang et al. [106]	2022	FFT	Self-supervised GCN	TUSZ	-
Lian et al. [111]	2022	Segment to 0.6 s	Spatial enhanced pattern feature +GCN	CHB-MIT	98.38
Rahmani et al. [107]	2023	FFT	Meta Learning +GAT	TUSZ	82.7
Jibon et al. [112]	2023	Stockwell Transformation	LGCN +DenseNet	CHB-MIT	98

of a generator and a discriminator. In long-term EEG recordings, epileptic seizure events are rare compared to non-seizure durations, posing a significant class imbalance challenge for seizure detection. GAN plays a vital role here, enabling the learning of deep representations without the need for abundant labeled training data [113]. The generator's objective is to produce samples resembling real epileptic signals, while the discriminator's role is to distinguish between generated samples and the original EEG records. By generating synthetic data, it becomes possible to argue the training dataset, particularly when actual epileptic data is limited, thereby enhancing the performance of seizure detection models. The detector proposed by You et al. [114], which focused on image generation and discrimination based behind-the-ear EEG by a GAN network with a convolutional layer (AnoGAN). This unsupervised learning model holds promise for practical applications.

Gao et al. used GAN to synthesize seizures-like samples to address the challenge of imbalanced data [115]. Additionally, the focal loss functions introduced in [103] and [104], which give more emphasis to the minority class, were designed to mitigate this issue and promote model performance in seizure detection. More detailed context about the class balance problem and data enhancement in epilepsy detection can be seen in Ref. [116].

5. Hybrid DL-based methods

In the preceding chapters, we extensively examined the application of individual classical deep learning methods in the field of epileptic seizure detection. In recent times, researchers have embarked on integrating diverse deep learning techniques to harness their respective strengths, thereby enhancing the performance and robustness of epileptic seizure detection. This chapter presents research on hybrid deep learning models, encompassing three primary directions: CNN-RNN hybrid models, CNN-Autoencoder hybrid models, and hybrid models based on attention mechanisms or Transformers. These investigations into hybrid deep learning models have garnered widespread attention in recent years owing to their potential to further improve performance in epileptic seizure detection tasks and demonstrate increased applicability in the intricate analysis of EEG signals.

5.1. CNN-RNNs

The combination of CNN-RNNs in deep learning hybrid models has gained significant popularity in epilepsy detection due to their unique capabilities. CNN excels in processing spatial features, while RNN specializes in capturing dependencies in time-series data. The integration of these two architectures allows the model to consider both the spatial and temporal characteristics of EEG signals, enabling a more comprehensive analysis and identification of epileptic events. Thodorof et al. devised a patient-specific detector that initially extracted EEG representations image-based leveraging FFT and then performed EEG classification tasks using a distinctive Recurrent convolutional network (CNN-LSTM) [117]. Similar to their work, [38] integrated 2D CNNs, 1-D CNNs and LSTM to extract potential features of EEG space-time. Nevertheless, they used a deep learning-based post-processor for epoch-based decision-making on the output of the deep network. A convolutional LSTM network (CLSTM) with 9 layers is designed in [118], whose output is obtained by dense layer, and satisfactory results are gained for both binary and triple classification of the Bonn dataset. A BiLSTM network, serving as a temporal classifier, performs binary classification on the feature vectors learned from CNN [119]. Validation of the model was conducted through leave-one-out cross-validation on the clinical EEG dataset obtained from Johns Hopkins Hospital (JHH), resulting in an average sensitivity rate of 91 %. In recent work, Natsu detected epilepsy using the attention-based Hierarchical Convolutional BiGRU (HCLA_CBiGRU), which achieved an impressive accuracy rate of 98.5 % across four

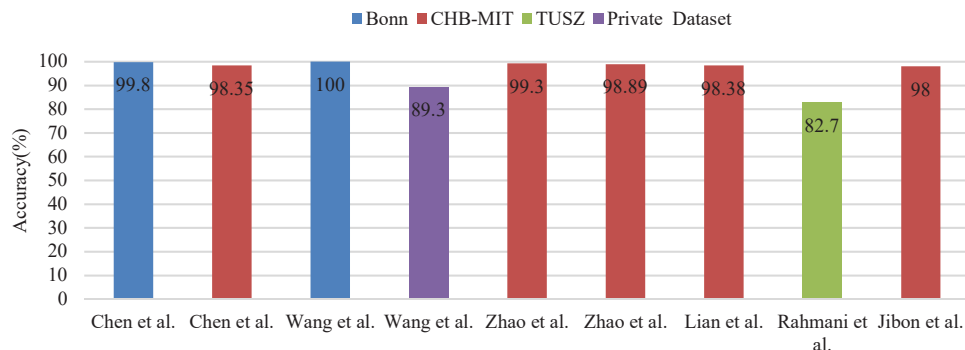


Fig. 9. Comparison of accuracy attained by different authors using GNNs models for identifying epilepsy.

combined EEG datasets [120]. Table 6 and Fig. 10 gives the literature information of detection models using CNN-RNNs in the last five years, respectively.

5.2. CNN-autoencoders

CNN possesses robust feature extraction capabilities, while autoencoders aim to minimize the discrepancy between input and generated output. Extensive evidence supports CNN-Autoencoder synergistic interaction, enhancing the model's capacity to capture intricate patterns in EEG data. Especially for the seizure recognition field, Wen et al. [129] developed an unsupervised feature representation model for epileptic seizure classification based on deep CNN and autoencoder. Experimental results demonstrate the model's ability to achieve favorable performance even with reduced feature dimensions. The FFT-converted EEG images were input into the autoencoder, and CNN was used for feature extraction to prepare the EEG signal classification task [90]. These two research studies diverge in terms of their network inputs: the former relies on EEG sequences, while the latter utilizes image data.

5.3. Hybrid architectures based on attention and transformer

In the context of the aforementioned hybrid models, both of which incorporate CNNs, it's worth noting that CNNs exhibit limitations in capturing global dependencies. In contrast, attention mechanisms exhibit superior capabilities, and it is non-trivial to recognize that constitute a pivotal technique within the domain of deep learning [130]. They empower models to learn relationships between distinct positions and capture contextual dependencies. In recent years, a plethora of research articles centered around attention mechanisms have emerged, with some studies utilizing attention as a foundational element to realize channel selection within multi-channel EEG signals. The authors led by Ye et al., introduced a multi-view deep learning framework called ChannelAtt [131]. This model seamlessly learns the spectrogram representation of EEG signals in an end-to-end manner. Additionally, a Global-based attention mechanism was chosen to compute scores that capture relevant channel information, thereby enhancing the detection performance. Isaev et al. devised two distinct deep learning networks on neonatal EEG, both rooted in Inception blocks for feature extraction [132]. The differentiation between them lies in the incorporation of an Attention mechanism in the second network, acknowledging the significance of Attention in the process. This study also delves into data balancing techniques and conducts model evaluations on the Duke and Helsinki datasets. Qiu et al. introduced an Attention-based ResNet-LSTM deep network capable of simultaneously extracting temporal and spatial information related to epileptic seizures [133]. The original EEG is accompanied by the computed backward difference signal as inputs to

the deep network. The difference layer serves the purpose of feature enhancement to unearth additional epileptic signal information, while the channel attention module adaptively amplifies seizure-related information. Similar to [121], an Attention-based CNN-BiGRU deep network for epileptic seizure identification was provided by Xu et al. [134].

The self-attention mechanism is a variant of the attention mechanism, while the Transformer is a neural network model based on self-attention. In [135], they introduced the transformer module to capture the global dependency information of EEG signals, yielding a robust classification outcome. In addition, [136,137] and [138] are all dedicated to the task of epileptic seizure detection while taking into account the spatial structure of EEG signals. In [136] a hybrid model featuring Graph ResNet and Attention mechanisms is employed. In the remaining two studies, GAT is integrated with Transformer in one, and GCN is combined with Transformer in the other, each tailored to address specific objectives. Table 7 and Fig. 11 present recent researches of seizure detection based on Attention or Transformer.

6. Integrating DL with traditional ML methods

The integration of deep learning (DL) techniques with traditional machine learning methods has formed a powerful framework. DL serves as the foundation for feature extraction, and its outputs are subsequently employed as inputs to traditional machine learning algorithms such as SVM, logistic regression (LR), and random forests. Due to its capability to automatically perform feature extraction by learning internal representations of data, DL overcomes limitations associated with certain traditional methods. In [34], the authors extracted several simple handcrafted features, which were then fed into Deep Belief Network (DBN). The DBN's outputs were subsequently input into three distinct machine learning classifiers (SVM, KNN, and LR). A stacked autoencoder learns feature representations for unannotated EEG recordings, and these learned features are then combined with the relevant labels of the original EEG dataset via a logistic regression classifier to detect anomalies in the EEG data [141].

A composite denoising AE (DAE) and sparse AE (SpAE), known as a denoising sparse autoencoder (DSpAE), is employed as a feature extractor. Its output is subsequently used as input for a logistic regression classifier. This typical model achieved the recognition of 100 % on the Bonn dataset of the two-class task [142]. The aforementioned research highlights the extensive utilization of logistic regression within contemporary deep learning investigations, consistently yielding commendable outcomes across diverse classification endeavors. Research endeavors such as [143–148] have corroborated the fusion of deep learning models with machine learning classification algorithm. Table 8 and Fig. 12 depicts methods proposed in recent 5 years that use

Table 6
Existing literature on using CNN-RNNs in recent 5 years.

Work	Year	Preprocessing technique	Method	Database	Accuracy (%)
Baloglu et al. [118]	2019	-	CNN-LSTM	Bonn	AE:100 ADE:98.33
Huang et al. [121]	2019	Segmented	Attention-based CNN-BiRNN	CHB-MIT	-
Roy et al. [122]	2019	Data selection	1D-Conv+GRU (ChronoNet)	TUSZ	-
Liang et al. [123]	2020	Segmented into 1 h	CNN-LSTM(LRCN)	CHB-MIT	99
Xu et al. [124]	2020	Normalization	1D CNN-LSTM	Epileptic Seizures Recognition EEG	99.39
Hussain et al. [125]	2021	FFT and DWT	1D CNN-LSTM	Freiburg	99.27
Craley et al. [119]	2021	High-pass and Low-pass filter	1D CNN-BiLSTM	Clinical Datasets of JHH	-
Kumar et al. [126]	2022	Band-pass filter	2D CNN-GRU	Bonn	100
Rahman et al. [127]	2022	Band-pass filter	1D CNN-GRU	TUSZ	97.78
Natu et al. [120]	2023	Merged datasets	HCLA CBiGRU	4 public Datasets	98.5
Rashed-Al-Mahfuz et al. [68]	2023	Normalization	1D-CNN with LSTM	CHB-MIT	99.25
Varli et al.[128]	2023	CWT STFT	2D CNN-LSTM	Bonn	98.97
				CHB-MIT	95.46
				Bern-Barcelona	96.23
					99.81
					93.77
					95.08

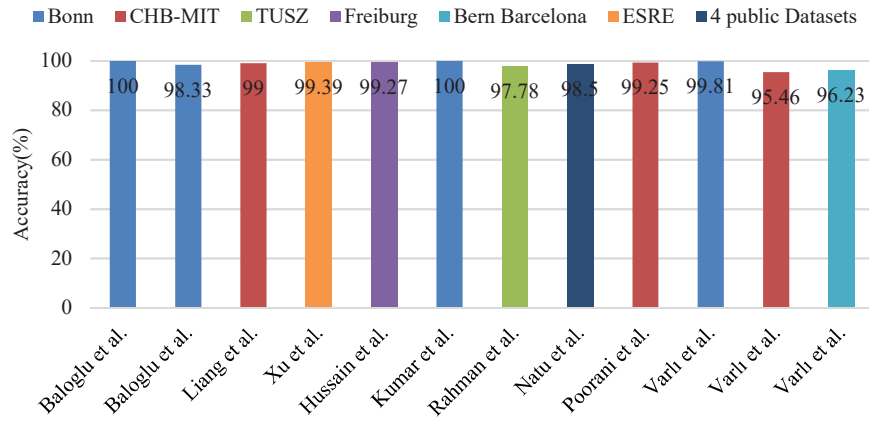


Fig. 10. Comparison of accuracy attained by different authors using CNN-RNNs models for identifying epilepsy.

Table 7

Existing literature on using Attention or Transformer in recent 5 years.

Work	Year	Preprocessing technique	Method	Database	Accuracy (%)
Yuan et al. [131]	2018	STFT	Multi-View Representation and attention mechanism	CHB-MIT	96.61
Isaev et al. [132]	2020	Notch, high-pass, and low-pass filter	Inception block and Attention	Duke Helsinki	-
Song et al. [135]	2021	Segmented, band-pass filter, and standardization	Transformer	2a of BCI	82.59
				2b of BCI	84.26
Dong et al. [136]	2022	Band-pass Filter & Z-score standardization	Attention-based Graph ResNet with focal loss	CHB-MIT	98.7
He et al. [139]	2022	Band-pass Filter & Z-score standardization	GAT + BiLSTM	CHB-MIT	98.52
				TUSZ	98.02
Qiu et al. [133]	2023	-	Attention-based ResNet-LSTM	Bonn	98.87
				Hauz Khas Delhi	100
Xu et al. [134]	2023	DWT	Feature extraction and Attention-based CNN-BiGRU	Freiburg	98.33
Zhao et al. [137]	2023	Band-pass filter	GAT + Transformer	CHB-MIT	98.3
Lian et al. [138]	2023	-	GCN + Transformer	CHB-MIT	98.43
Si et al. [140]	2023	Downsampled & Band-pass filter	Convolutional Block Attention Module	SWEC-ETHZ	-
				TJU-HH	-

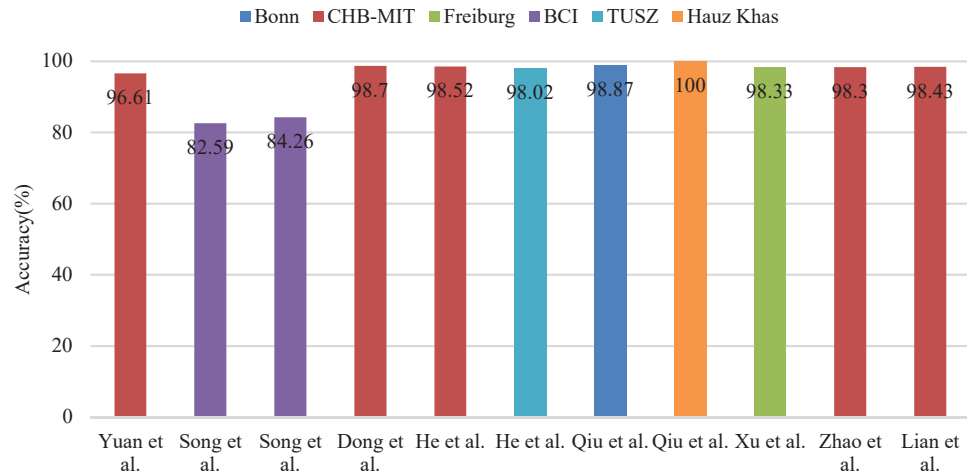


Fig. 11. Comparison of accuracy attained by different authors using Attention or Transformer models for identifying epilepsy.

DL techniques combined with traditional ML methods for epileptic seizure detection.

7. Discussion and challenges

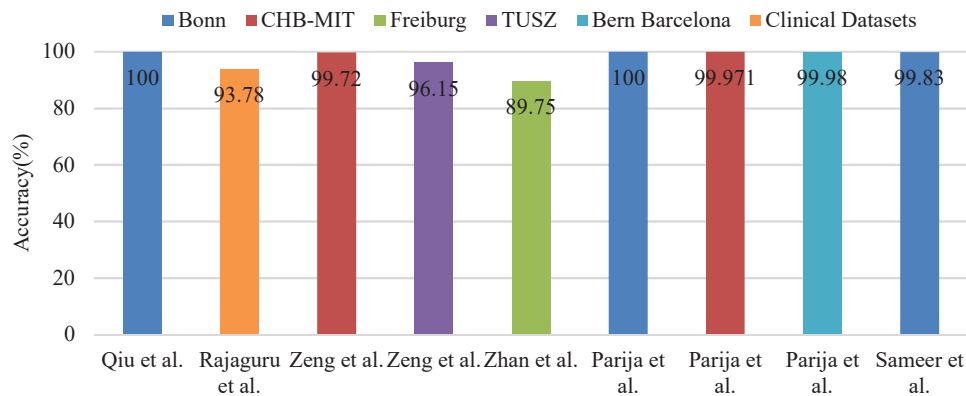
Deep learning techniques have found notable success on epileptic seizure detection. Through a retrospective examination of the developmental trajectory within the realm of seizure detection, it becomes

unequivocally evident that an automatic seizure detection system should possess the capability to accurately discriminate between epileptic and non-epileptic EEG signals. We have meticulously conducted a systematic analysis of methodologies grounded in deep learning technologies, delving deeply into the intricacies of commonly employed neural network architectures. Firstly, methods based on classical deep learning techniques were investigated, and commonly used neural network architectures were deeply analyzed, including

Table 8

Existing literature on using integrating DL with Traditional ML Methods in recent 5 years.

Work	Year	Preprocessing technique	Method	Database	Accuracy (%)
Qiu et al. [142]	2018	Z-Score standardization	DSPAE +LR	Bonn	100
Rajaguru et al. [143]	2018	-	Multilayer Autoencoder & EM-PCA+ Genetic Algorithm	Clinical Datasets	93.78&93.92
Ansari et al. [144]	2019	Segmented	Feature extraction + CNN + Random Forest	Neonatal Clinical Datasets	-
Zeng et al. [145]	2020	Band stop filter and Z-Score standardization	Time and frequency domain features +HGCN +Tree Classification	CHB-MIT	99.72
Zhan et al. [146]	2020	-	Feature extraction + DCNN + Multiview Fuzzy Clustering	TUSZ	96.15
Parija et al. [147]	2021	Segmented and band pass filter	MVKRVFLN+ DLSTM	Freiburg	89.75
				Bonn	100
				CHB-MIT	99.971
				Bern Barcelona	99.98
Sameer et al. [148]	2022	-	CNN+ LR	Bonn	99.83

**Fig. 12.** The Comparison chart of accuracy attained by different authors using integrating DL with Traditional ML methods for identifying epilepsy.

CNNs, RNNs, DBNs, Autoencoders, GCNs, and GANs. Emphasis was particularly placed on the critical roles of CNNs, RNNs, and GCNs within this context. Secondly, the emerging field of hybrid deep learning models was explored, recognizing their future trends in epilepsy detection. We meticulously dissected existing hybrid models, elucidating their distinctive combinations and conducted a comparative evaluation of their performance across diverse datasets. It is believed that hybrid deep learning methods will continue to play an increasingly significant role in epileptic seizure detection in the future. Thirdly, the synergistic integration of deep learning techniques with traditional machine learning methods was deeply studied. Notably, we highlighted the role of deep learning as a feature extractor, coupled with traditional machine learning classification algorithms. Importantly, the remarkable effectiveness of logistic regression when combined with deep learning representations was presented. These sophisticated deep learning approaches furnish us with potent tools and strategies to propel advancements in the field of seizure detection.

Compared with traditional machine learning methods, deep learning methods have their own unique advantages, such as automatic feature extraction, good at processing large-scale data and higher detection accuracy. However, it is important to note that while deep learning has shown remarkable potential in advancing seizure detection, several challenges and limitations remain which can be summarized as follows:

- **Data Quality and Imbalance:** Obtaining and preparing high-quality EEG data for training deep learning models remains a primary concern. There is also a class imbalance issue, where seizure events are relatively rare compared to non-seizure segments. It is possible to increase the diversity and amount of training data by implementing data enhancement techniques such as using SMOTE or GANs to generate synthetic data. Techniques such as resampling, class weighting, or cost-sensitive learning are applied to mitigate the effects of class imbalances. In addition, collaboration between research

institutions is encouraged to establish standardized EEG datasets that would be shared within the scientific community, thereby improving data quality and availability.

- **Interpretability:** The interpretability of deep learning models is an ongoing concern. Like CNNs and RNNs, while highly effective, are often considered 'black-box' models, making it challenging to understand the features they extract and their decision-making processes. Enhancing the interpretability and credibility of these models is a pressing research direction. Insights into the model decision process can be provided by combining techniques like Layer-wise Relevance Propagation (LRP), Grad-CAM, or SHAP values. Moreover, applying simpler models or hybrid approaches that combine deep learning with more explainable models can balance accuracy and transparency.
- **Robustness and Clinical validation:** The robustness of deep learning models to variations in EEG data across different patients and recording conditions is an area that requires further attention. A model that performs well on one dataset may not effectively generalize to different clinical environments, highlighting the need for more adaptable and transferable models. Cross-validation of multiple datasets from different sources can enhance the robustness of the model, and the proposed model could be generalized in different clinical settings to promote the development of domain adaptation techniques. More extensive validation studies in collaboration with clinical partners would be also critical to translating the model into practical applications.
- **Limited Computational Resources:** Deploying deep learning models in real applications presents practical challenges related to computational resources and model optimization. Computational demands could be reduced by focusing on developing lightweight models. Moreover, exploring cloud-based solutions or edge computing could offer practical avenues for deploying these models in resource-constrained environments.

Addressing these challenges and limitations is crucial for the continued advancement of deep learning-based epileptic seizure detection and its successful translation into clinical practice.

8. Conclusion

In this comprehensive review, we have explored the diverse landscape of epileptic seizure detection through the lens of deep learning. Our analysis has covered three fundamental aspects: the application of canonical DL methods, the emergence of hybrid DL models, and the integration of DL with traditional ML techniques. However, our exploration also revealed challenges and limitations, including data scarcity, interpretability issues, and the need for rigorous clinical validation. Looking ahead, the future of EEG-based seizure detection lies in addressing these challenges. The establishment of large-scale, high-quality EEG datasets is paramount. Research into techniques for model interpretability, feature visualization, and model transparency will contribute to a better understanding and trust in automatic detection systems. Fostering collaboration between the deep learning and clinical communities, coupled with extensive clinical validation to demonstrate the accuracy of detection models, is of utmost importance. Additionally, the development of lightweight models and cloud-based solutions is imperative to enhance practicality. In conclusion, this article primarily serves as a comprehensive synthesis of the work done in automatic epileptic seizure detection using deep learning techniques applied to EEG signals, offering researchers a fresh perspective on this research domain and avenues for future advancements.

CRediT authorship contribution statement

Yankai Yang: Conceptualization. **Zengqian Deng:** Methodology. **Kuiting Yan:** Conceptualization. **Jie Xu:** Writing – original draft. **Shasha Yuan:** Writing – review & editing. **Juan Wang:** Conceptualization. **Jin-Xing Liu:** Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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